WEEK 3

Churn Prediction & Retention Analysis Report

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Final Dataset: production_ready_dataset_v3_validated.xlsx

Total Records: 8,558

1. Introduction

Churn—defined as a student dropping out or withdrawing from a program—has a direct impact on program outcomes, resource planning, operational continuity, and overall student satisfaction. Predicting churn early enables proactive outreach, reduces preventable drop-offs, and ensures higher completion rates.

Objective

The goal of this analysis is to build a robust and interpretable churn prediction pipeline that:

- Identifies high-risk students early enough for targeted intervention.
- Quantifies the key drivers of churn to understand *who* is likely to drop off and *why*.
- Supports data-driven retention strategies for operational teams.

This analysis uses the validated dataset **production_ready_dataset_v3_validated.xlsx**, focusing on binary churn prediction:

- **1 = Dropped Out / Withdraw**
- **0 = All other outcomes**

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2. Data Preparation

2.1 Raw Data and Target Definition

- **Source Dataset:** production_ready_dataset_v3_validated.xlsx
- **Total Rows:** ≈ 8,500 records after cleaning
- **Target Variable (churn):**
  - 1 if status_description ∈ {Dropped Out, Withdraw}
  - 0 otherwise
- **Class Distribution:**
  - Non-churn: **~91.8%**
  - Churn: **~8.2%**

This level of imbalance is manageable for boosting models and class-weighted linear models.

2.2 Data Cleaning and Leakage Handling

Columns Removed

Identifiers:

- opportunity_id, opportunity_name, first_name

Raw Date Fields:

- opportunity_end_date, opportunity_start_date
- apply_date, entry_created_at, date_of_birth

Leakage-Prone Fields (post-labeling):

- status_description, status_code

Rationale

- Identifier fields provide no predictive signal.
- Raw dates are replaced with **engineered time-delta features**, which are more meaningful for churn modeling.
- status_* fields encode the final outcome directly and must be removed after defining the target variable.

2.3 Final Feature Set

Numeric Features

- age_years
- signup_month, signup_year
- engagement_lag_days
- opportunity_duration_days
- days_before_start
- flag_engagement_inversion

- flag_days_before_start_extreme
- log_opportunity_duration
- applied_after_start
- engagement_lag_days_fixed
- gender_encoded

Categorical Features

- opportunity_category
- country
- institution_name
- current_intended_major
- engagement_lag_bucket

All categorical variables were cast to **string** before encoding to avoid mixed-type issues.

2.4 Missing Value Handling and Scaling

Numeric:

- **Median imputation** (SimpleImputer)
- **Standard scaling** (StandardScaler)

Categorical:

- **Most frequent imputation**
- **One-Hot Encoding** (OneHotEncoder(handle_unknown="ignore"))

All preprocessing steps were encapsulated in a unified **Pipeline + ColumnTransformer** to ensure reproducibility and prevent data leakage.

2.5 Train/Test Split

- **Split:** 80% training, 20% testing
- **Method:** Stratified split to preserve original churn distribution

- **Resulting sets:**
  - Train: **6,846 records**
  - Test: **1,712 records**

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3. Exploratory Data Analysis (EDA)

3.1 Descriptive Statistics

Key observations:

- **Age**: Concentrated around early-20s with some outliers retained due to realistic demographic variation.
- **Timing Features**:
  - engagement_lag_days exhibited high variance.
  - days_before_start showed extreme early- and late-application behavior.
- **Churn Rate:** ~8–9%, indicating moderate imbalance.

3.2 Visualizations

Overall Churn Distribution

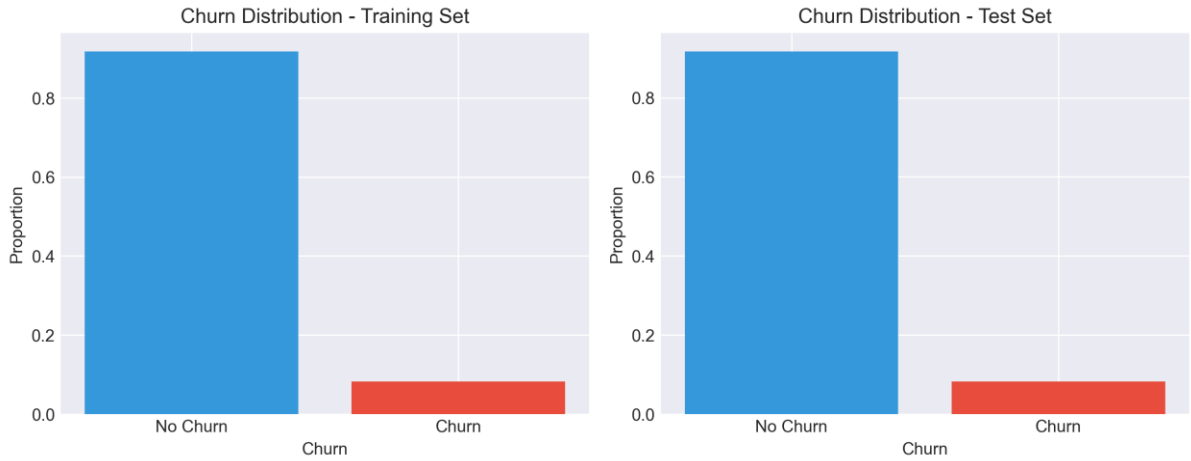

Figure 1. *Churn vs. Non-Churn Frequency*

Churn by Categorical Variables

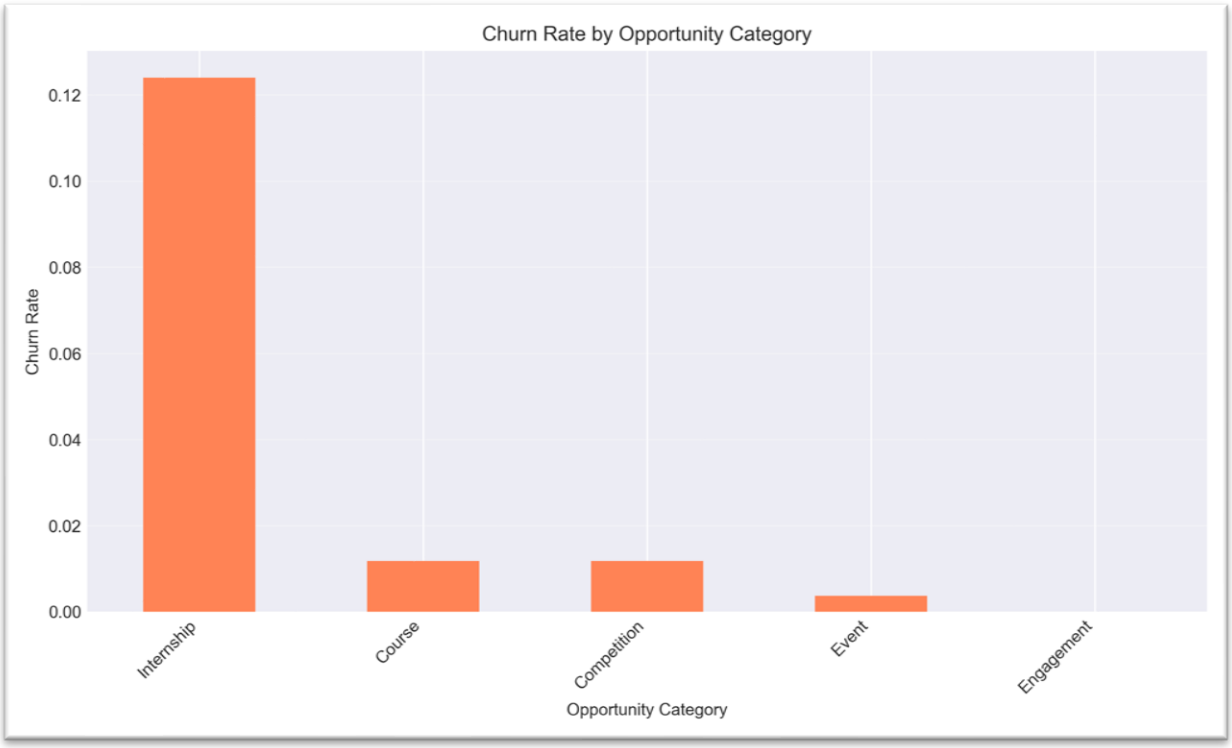

Figure 2. *Churn Rate by Opportunity Category*

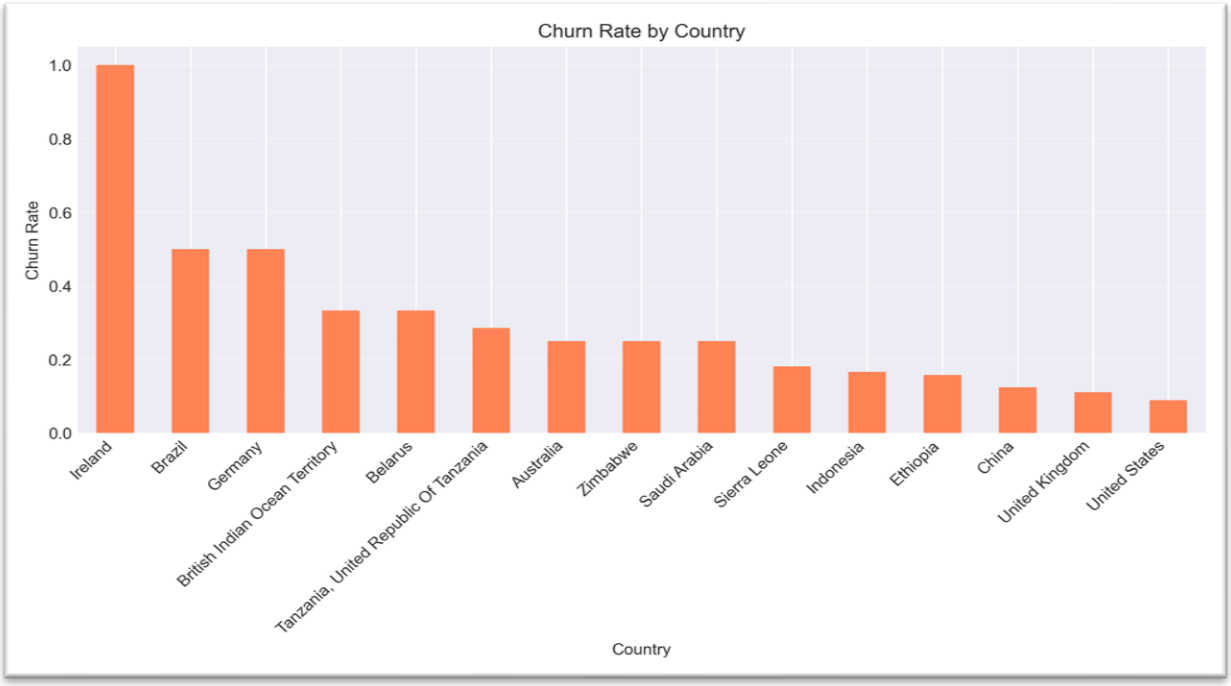

Figure 3. *Churn Rate by Country*

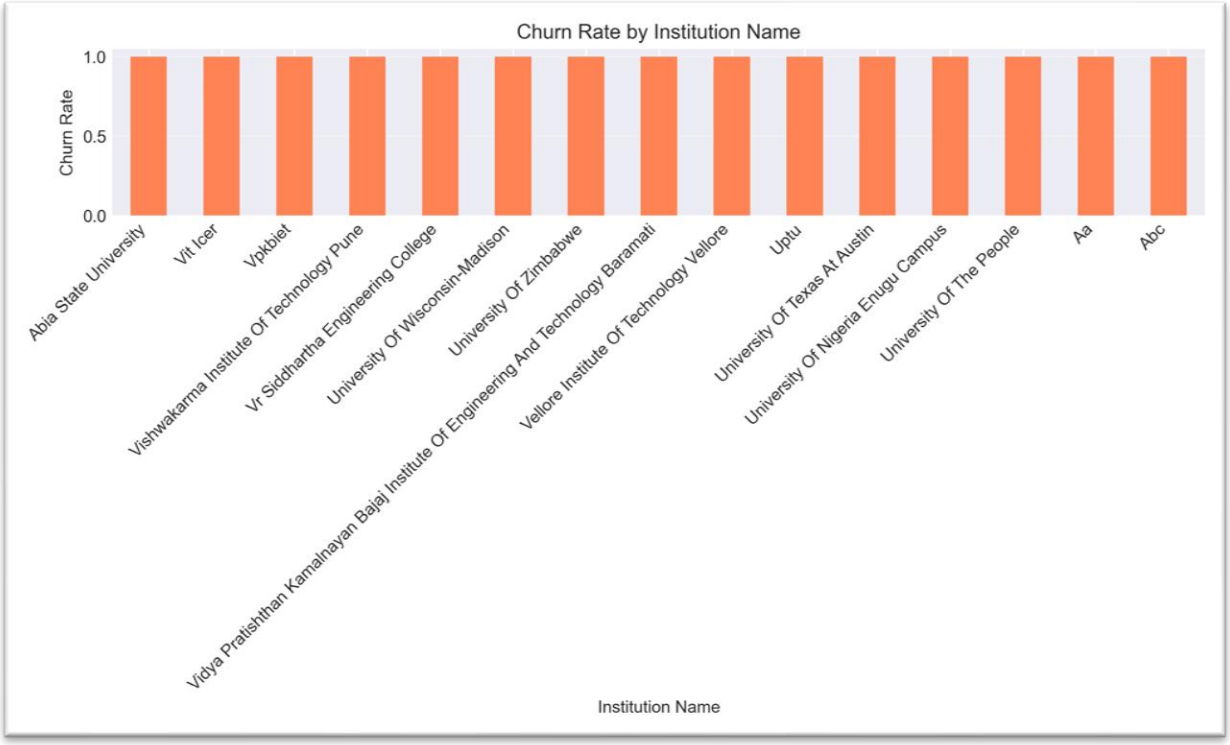

Figure 4. *Churn Rate by Institution (Top N)*

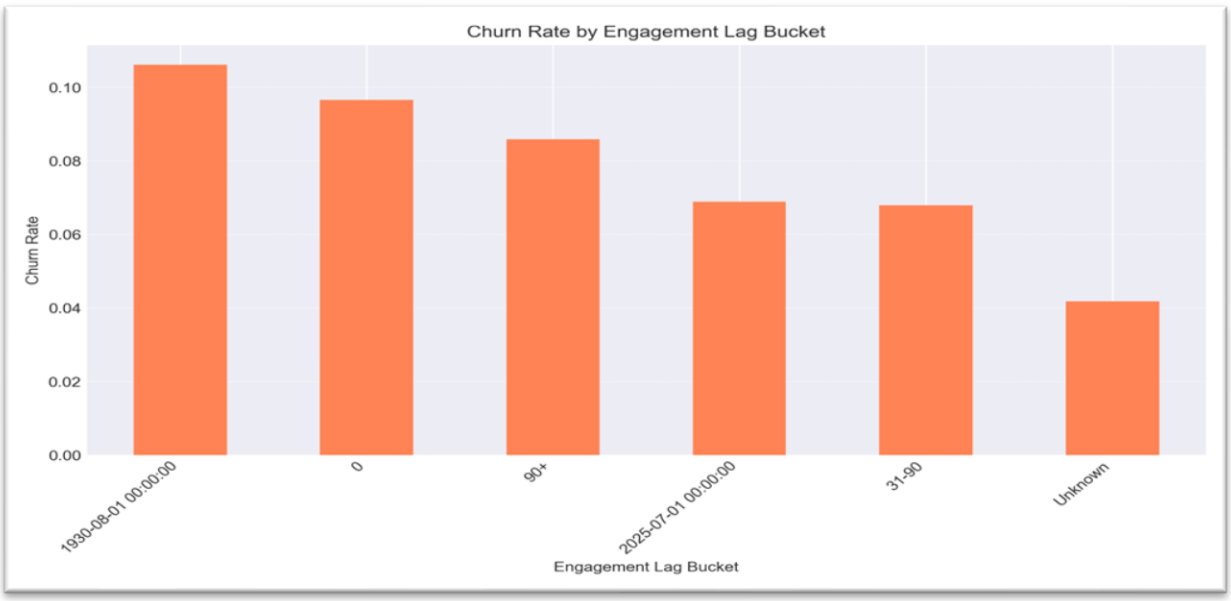

Figure 5. *Churn Rate by Engagement Lag Bucket*

Numeric Feature Distribution

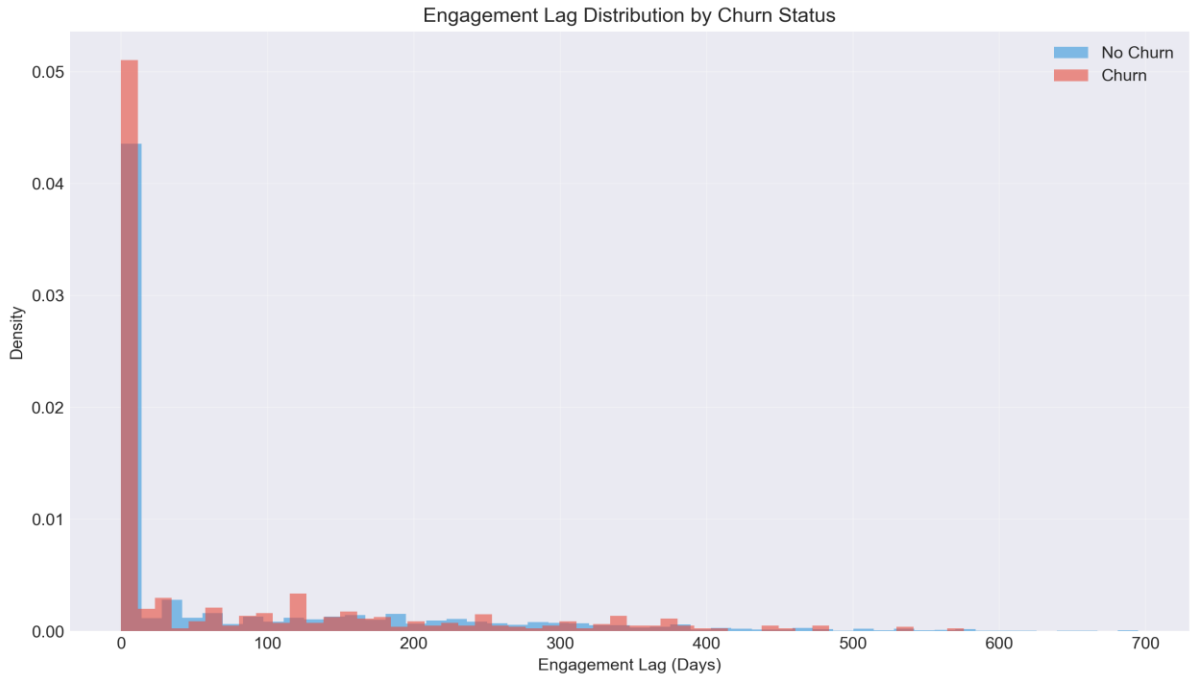

Figure 6. *Engagement Lag by Churn Status*

3.3 Patterns and Trends

Key insights:

- **Opportunity categories** vary significantly in churn behavior.
- **Countries and institutions** exhibit different churn rates—indicating contextual or environmental factors.
- **Long engagement lags** strongly correlate with churn risk.
- **Extreme values of days_before_start** (too early or too late) show higher drop-off rates.

These observations guided the modeling and feature engineering steps.

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4. Predictive Modeling

4.1 Models Evaluated

All models used the *same preprocessing pipeline* for fairness.

1. **Logistic Regression**
    - Good for interpretability
    - Handles imbalance via class weights
2. **Random Forest**
    - Captures non-linear patterns
    - Robust to outliers
3. **Gradient Boosting**
    - Strong tabular data performance
    - Excellent ranking ability

4.2 Training Procedure

- Models trained on **80% stratified training data**
- Preprocessing included inside the pipeline
- Imbalance addressed via class weights and boosting logic

4.3 Base Model Performance (Threshold = 0.5)

Model	ROC-AUC	PR-AUC	Precision	Recall	F1-Score
Gradient Boosting	**0.927**	**0.566**	0.678	0.284	0.400
Random Forest	0.901	0.498	0.604	0.411	0.489
Logistic Regression	0.878	0.383	0.268	**0.801**	0.402

Observations:

- **Gradient Boosting** achieved the **best ROC-AUC and PR-AUC**.
- Logistic Regression provides high recall but low precision.
- Random Forest balances both but underperforms boosting.

4.4 Threshold Tuning for High-Precision Scenarios

To support an operational mode where **false positives must be minimized**:

- Target precision chosen: **≥ 0.80**
- Gradient Boosting achieved:
  - **Precision:** 0.821
  - **Recall:** 0.163
  - **F1-Score:** 0.272
  - **Threshold:** ≈ 0.75
  - **Confusion Matrix:**
    - [1566,5,
      118,23]

This setting captures fewer churners but ensures the flagged ones are highly likely to churn.

4.5 Visual Performance Summaries

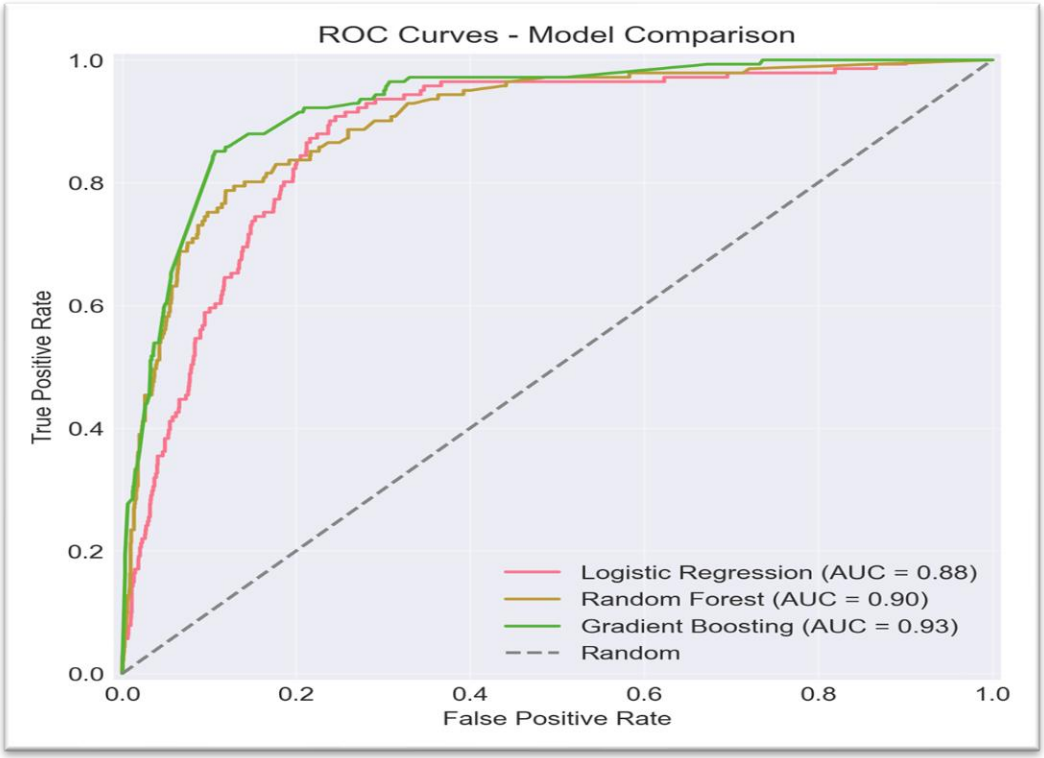

Figure 7. *ROC Curves for All Models*

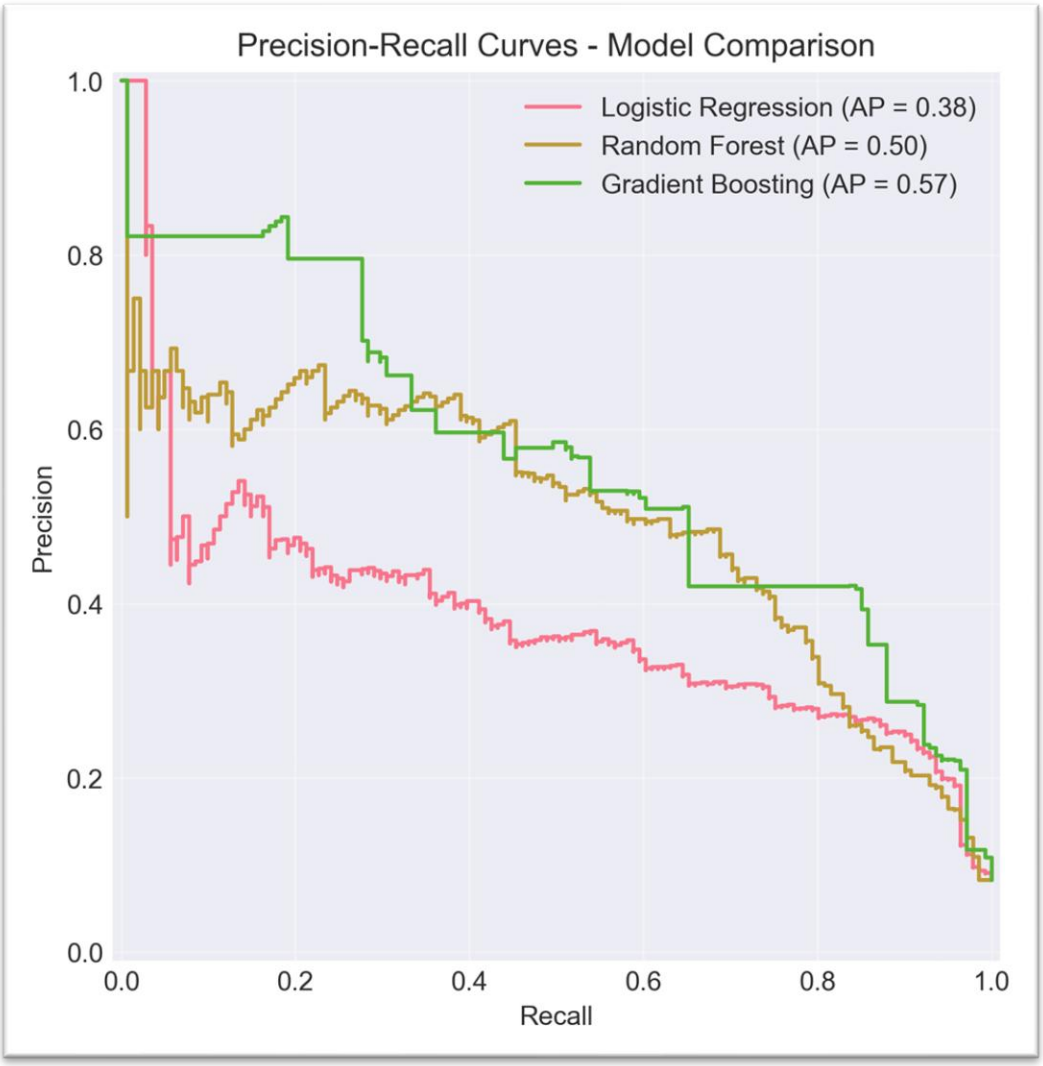

Figure 8. *Precision–Recall Curves for All Models*

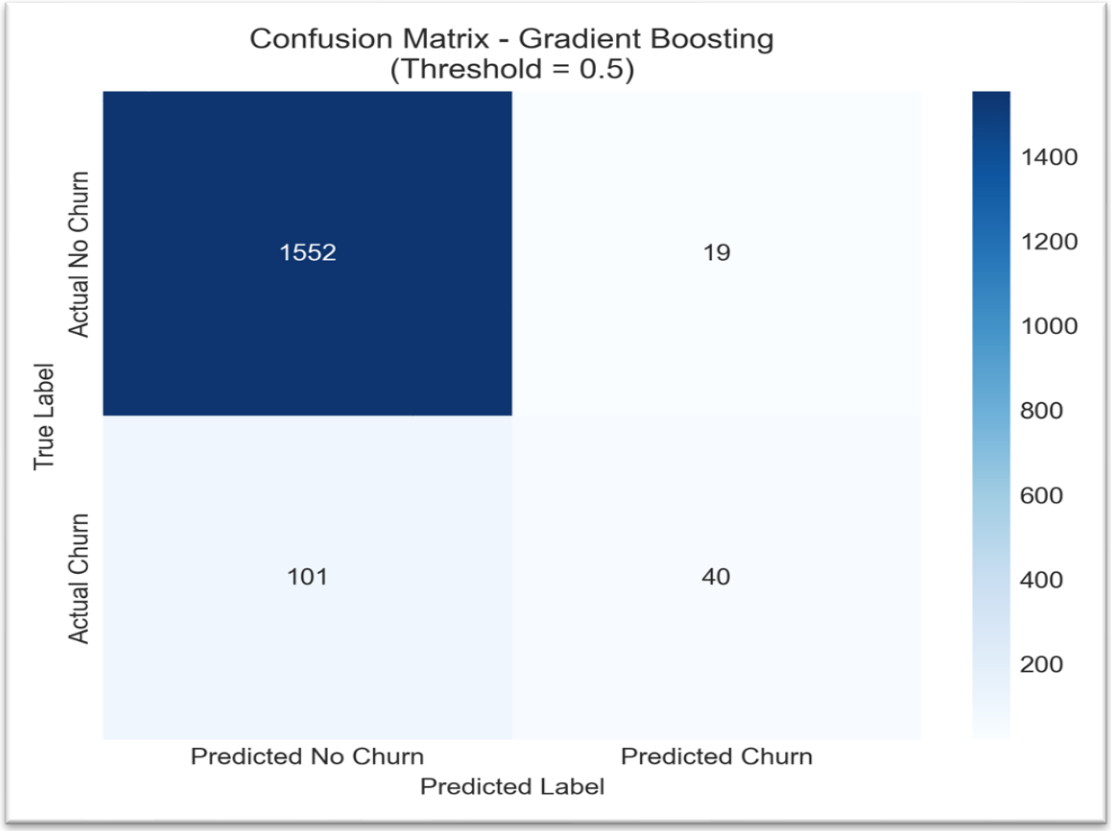

Figure 9. *Confusion Matrix – Gradient Boosting*

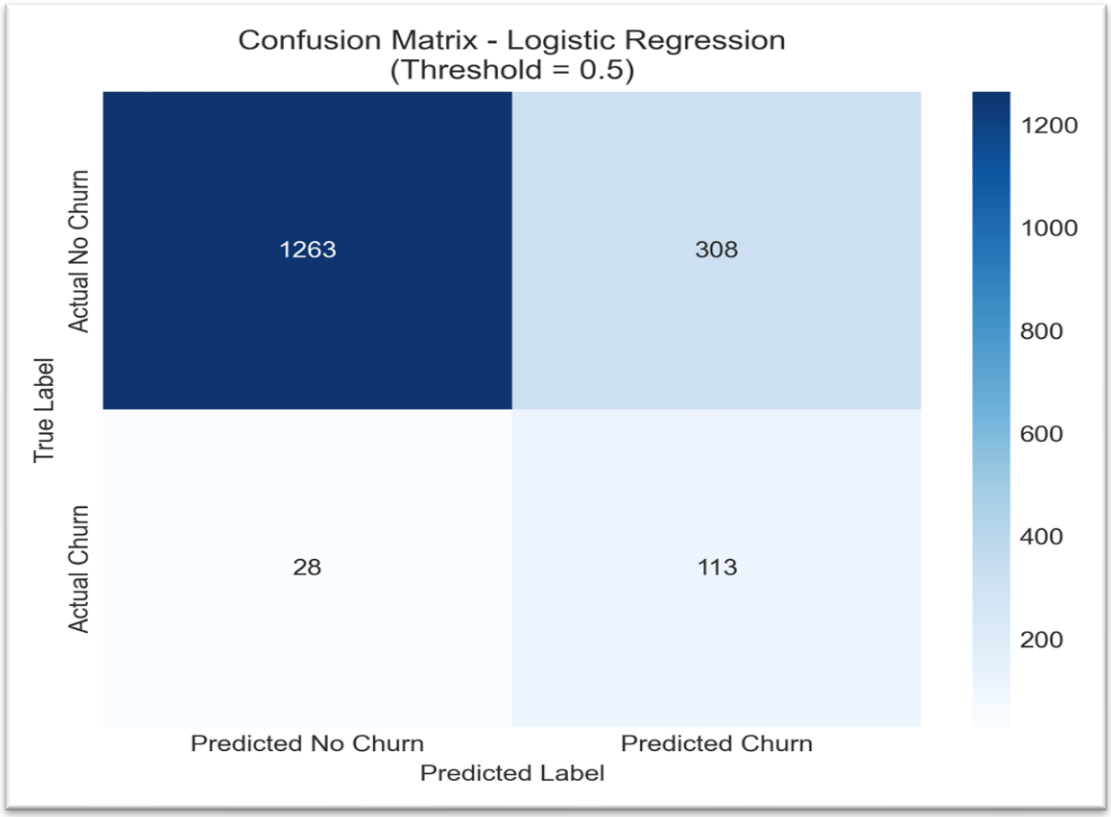

Figure 10. *Confusion Matrix – Logistic Regression*

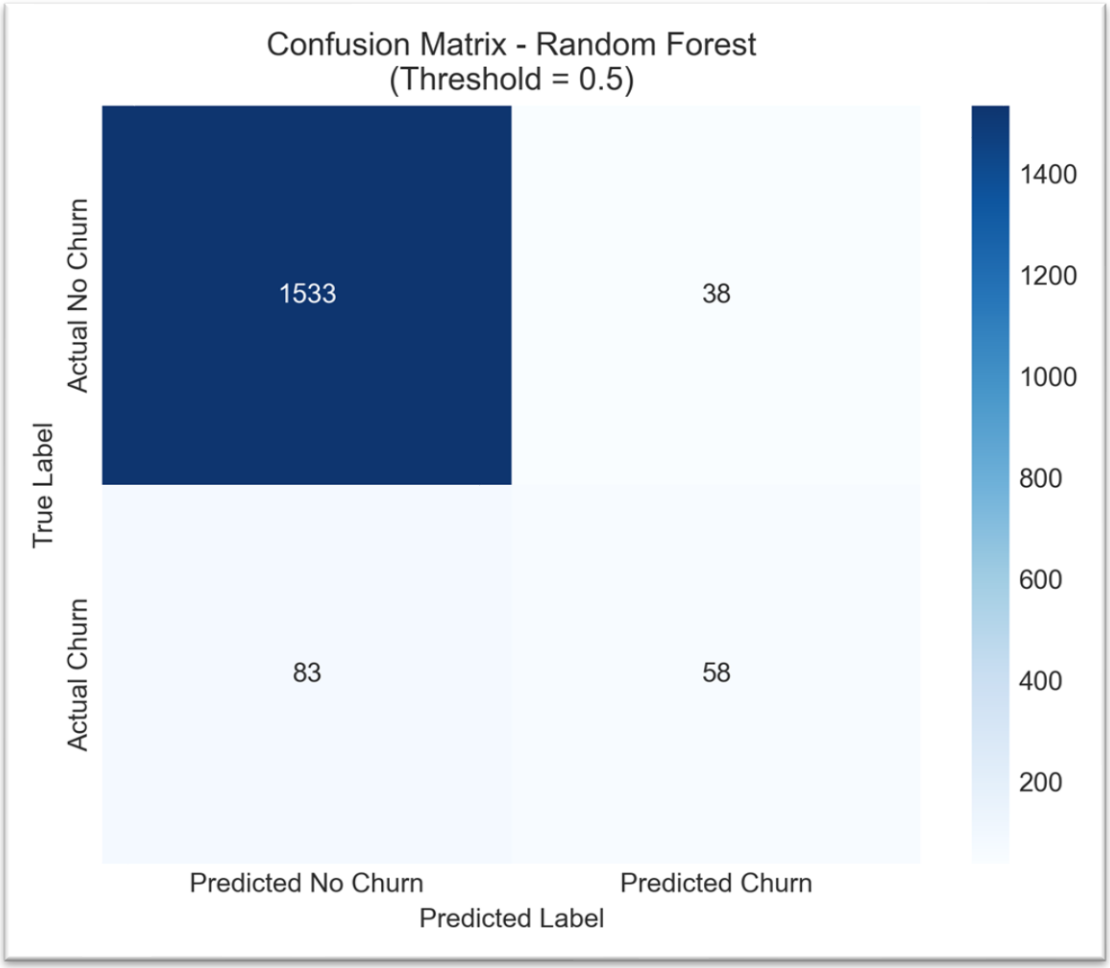

Figure 11. *Confusion Matrix – Random Forest*

5. Churn Analysis

5.1 Feature Importance

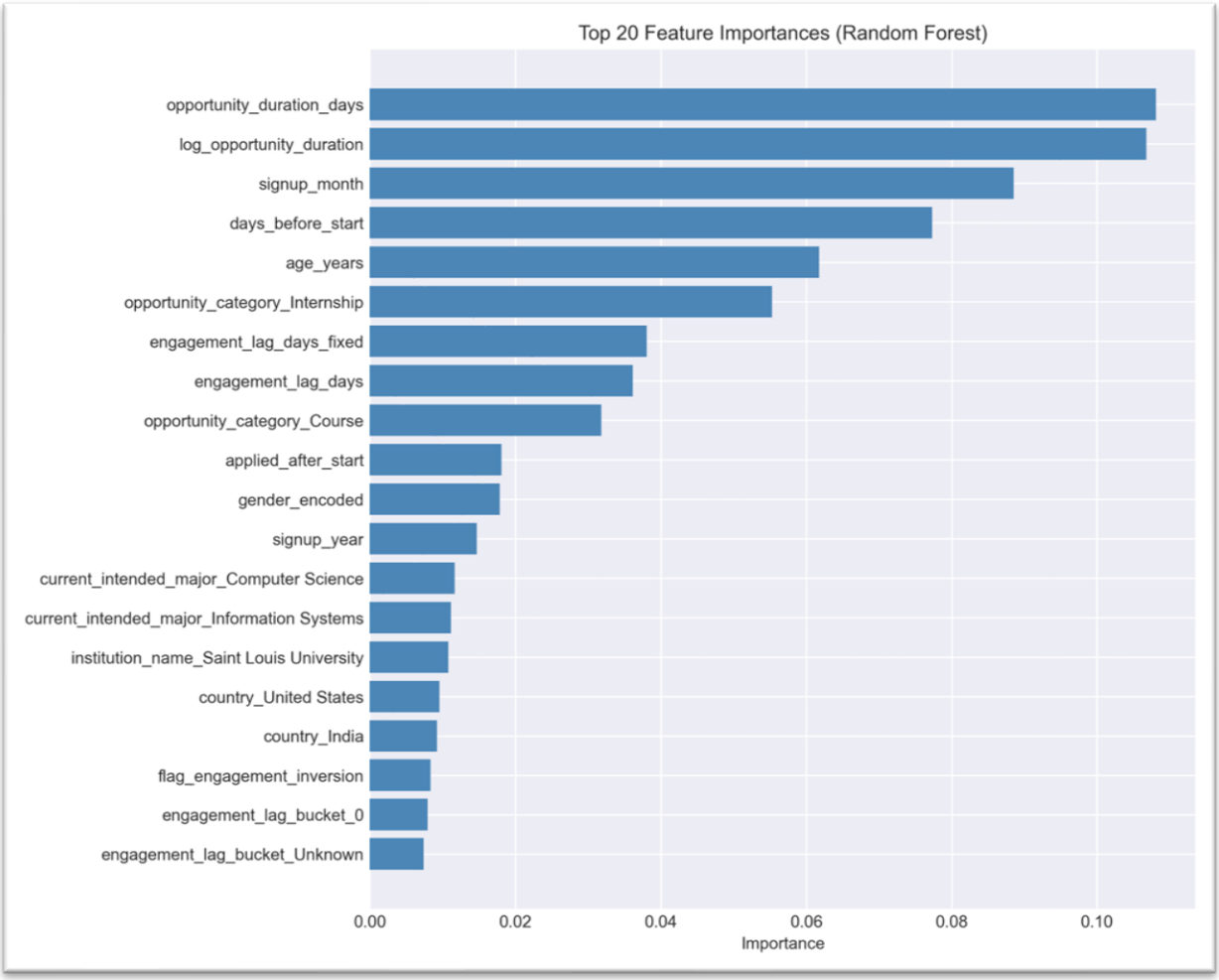

Figure 12. *Top 20 Feature Importances (Random Forest)*

Top predictors include:

- **Engagement Lag Features**
- **Opportunity Category**
- **Program Duration Variables**
- **Days Before Start**
- **Country & Institution**

These factors heavily influence the model's understanding of churn risk.

5.2 Churn by Segment

Segment-wise patterns show:

- Certain **program categories** consistently have higher churn.
- Some **countries and institutions** exhibit structural retention issues.
- **High engagement lag** is a powerful indicator of drop-off.

5.3 Impact Interpretation

- Longer engagement wait times often signal disengagement.
- Late applications reflect lower readiness; early applications require ongoing nurturing.
- Program structure (duration, category) plays a major role in retention outcomes.

6. Recommendations

6.1 Early Warning & Risk Scoring

Adopt the **Gradient Boosting model** for operational use:

- Generate continuous risk scores.
- Use higher precision threshold (≥ 0.80) based on team capacity.

6.2 Targeted Support Interventions

- **High-Risk Students:** Personalized mentor check-ins, onboarding calls.
- **High-Lag Students:** Accelerated first-touch communication.
- **Specific Opportunity Categories:** Additional orientation & coaching.
- **Geographic Hotspots:** Localized support strategies.

6.3 Program Design Improvements

- Review program categories with consistently high churn.
- Incorporate qualitative surveys from churned students.
- Map potential friction points early in the program journey.

7. Conclusion & Future Work

7.1 Summary

- A complete churn prediction pipeline was built and evaluated.
- **Gradient Boosting** delivered the strongest performance (ROC-AUC ≈ 0.93).
- Churn is strongly influenced by engagement timing, program design, and geography.
- Operationalizing this model can significantly improve early interventions.

7.2 Future Work

- Add behavioral engagement signals (logins, submissions).
- Conduct SHAP/LIME explainability analysis.
- Build dashboards and automate weekly churn scoring.
- Develop causal models to identify root-cause drivers of churn.